

## Questions

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1. (a) Show that

$$3 \cosh x - \sinh x = e^x + 2e^{-x}$$

[2]

(b) Hence solve the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 6(3 \cosh x - \sinh x)$$

given that, when  $x = 0$ ,  $y = 1$  and  $\frac{dy}{dx} = 1$ .

[10]

### Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

we have

$$\begin{aligned} 3 \cosh x - \sinh x &= 3 \left( \frac{e^x + e^{-x}}{2} \right) - \left( \frac{e^x - e^{-x}}{2} \right) \\ &= \frac{3e^x + 3e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^x + 4e^{-x}}{2} \\ &= e^x + 2e^{-x} \end{aligned}$$

Hence

$$3 \cosh x - \sinh x = e^x + 2e^{-x}$$

(b) From part (a),

$$6(3 \cosh x - \sinh x) = 6(e^x + 2e^{-x}) = 6e^x + 12e^{-x}$$

so the differential equation becomes

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 6e^x + 12e^{-x}$$

First solve the homogeneous equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$$

The auxilliary equation is

$$\lambda^2 - \lambda - 2 = 0$$

Factorising,

$$\begin{aligned} \lambda^2 - \lambda - 2 &= 0 \\ (\lambda - 2)(\lambda + 1) &= 0 \end{aligned}$$

So the roots are  $\lambda = 2$  and  $\lambda = -1$ . Therefore the complementary function is

$$y_c = Ae^{2x} + Be^{-x}$$

Now find a particular integral. Since the right-hand side is  $6e^x + 12e^{-x}$ , take

$$y_p = Ce^x + Dxe^{-x}$$

We use  $Dxe^{-x}$  rather than  $De^{-x}$  because  $e^{-x}$  is already part of the complementary function.

For the  $Ce^x$  term,

$$\frac{d}{dx}(Ce^x) = Ce^x, \quad \frac{d^2}{dx^2}(Ce^x) = Ce^x$$

Substituting into the left-hand side,

$$\begin{aligned} \frac{d^2}{dx^2}(Ce^x) - \frac{d}{dx}(Ce^x) - 2(Ce^x) &= Ce^x - Ce^x - 2Ce^x \\ &= -2Ce^x \end{aligned}$$

This must equal  $6e^x$ , so

$$-2C = 6$$

Hence

$$C = -3$$

For the  $Dxe^{-x}$  term,

$$\begin{aligned} y &= Dxe^{-x} \\ \frac{dy}{dx} &= D \frac{d}{dx}(xe^{-x}) \\ &= D(e^{-x} - xe^{-x}) \\ &= D(1 - x)e^{-x} \end{aligned}$$

Differentiate again:

$$\begin{aligned} \frac{d^2y}{dx^2} &= D \frac{d}{dx}((1 - x)e^{-x}) \\ &= D((-1)e^{-x} + (1 - x)(-e^{-x})) \\ &= D(x - 2)e^{-x} \end{aligned}$$

Substituting into the left-hand side,

$$\begin{aligned} \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y &= D(x - 2)e^{-x} - D(1 - x)e^{-x} - 2Dxe^{-x} \\ &= D(x - 2 - 1 + x - 2x)e^{-x} \\ &= -3De^{-x} \end{aligned}$$

This must equal  $12e^{-x}$ , so

$$-3D = 12$$

Hence

$$D = -4$$

Therefore a particular integral is

$$y_p = -3e^x - 4xe^{-x}$$

and the general solution is

$$y = Ae^{2x} + Be^{-x} - 3e^x - 4xe^{-x}$$

Now use the initial conditions.

When  $x = 0$ ,  $y = 1$ :

$$\begin{aligned} 1 &= Ae^0 + Be^0 - 3e^0 - 4(0)e^0 \\ &= A + B - 3 \end{aligned}$$

So

$$A + B = 4$$

Differentiate the general solution:

$$\begin{aligned} \frac{dy}{dx} &= 2Ae^{2x} - Be^{-x} - 3e^x - 4 \frac{d}{dx}(xe^{-x}) \\ &= 2Ae^{2x} - Be^{-x} - 3e^x - 4(1 - x)e^{-x} \end{aligned}$$

When  $x = 0$ ,  $\frac{dy}{dx} = 1$ :

$$\begin{aligned} 1 &= 2Ae^0 - Be^0 - 3e^0 - 4(1-0)e^0 \\ &= 2A - B - 3 - 4 \\ &= 2A - B - 7 \end{aligned}$$

So

$$2A - B = 8$$

Solve the simultaneous equations

$$\begin{aligned} A + B &= 4 \\ 2A - B &= 8 \end{aligned}$$

Adding gives

$$3A = 12$$

so

$$A = 4$$

Then

$$B = 4 - A = 0$$

Substituting back,

$$y = 4e^{2x} - 3e^x - 4xe^{-x}$$

Therefore the required solution is

$$y = 4e^{2x} - 3e^x - 4xe^{-x}$$

2. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 8e^{-2t} \sin t$$

given that, when  $t = 0$ ,  $y = 1$  and  $\frac{dy}{dt} = 2$ .

[10]

### Solution

For the homogeneous part,

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 13 = 0$$

So

$$\lambda = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$$

Hence

$$y_c = e^{-2t} (A \cos 3t + B \sin 3t)$$

For a particular integral, try

$$y_p = e^{-2t} (a \sin t + b \cos t)$$

Then

$$\begin{aligned} \frac{dy_p}{dt} &= \frac{d}{dt} [e^{-2t} (a \sin t + b \cos t)] \\ &= -2e^{-2t} (a \sin t + b \cos t) + e^{-2t} (a \cos t - b \sin t) \\ &= e^{-2t} [(-b - 2a) \sin t + (a - 2b) \cos t] \end{aligned}$$

Also,

$$\begin{aligned} \frac{d^2y_p}{dt^2} &= \frac{d}{dt} \{e^{-2t} [(-b - 2a) \sin t + (a - 2b) \cos t]\} \\ &= -2e^{-2t} [(-b - 2a) \sin t + (a - 2b) \cos t] \\ &\quad + e^{-2t} [(-b - 2a) \cos t - (a - 2b) \sin t] \\ &= e^{-2t} [(3a + 4b) \sin t + (-4a + 3b) \cos t] \end{aligned}$$

Substituting into

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 8e^{-2t} \sin t$$

gives

$$\begin{aligned} &e^{-2t} [(3a + 4b) \sin t + (-4a + 3b) \cos t] \\ &\quad + 4e^{-2t} [(-b - 2a) \sin t + (a - 2b) \cos t] \\ &\quad + 13e^{-2t} (a \sin t + b \cos t) \\ &= 8e^{-2t} \sin t \end{aligned}$$

So

$$e^{-2t} (8a \sin t + 8b \cos t) = 8e^{-2t} \sin t$$

Hence

$$8a \sin t + 8b \cos t = 8 \sin t$$

Comparing coefficients,

$$8a = 8, \quad 8b = 0$$

So

$$a = 1, \quad b = 0$$

Therefore

$$y_p = e^{-2t} \sin t$$

Thus the general solution is

$$y = e^{-2t} (A \cos 3t + B \sin 3t) + e^{-2t} \sin t$$

So

$$y = e^{-2t} (A \cos 3t + B \sin 3t + \sin t)$$

Now use the initial conditions

Given  $y(0) = 1$ ,

$$1 = e^0 (A \cos 0 + B \sin 0 + \sin 0)$$

So

$$1 = A$$

Hence

$$A = 1$$

Now differentiate

$$\begin{aligned} \frac{dy}{dt} &= \frac{d}{dt} [e^{-2t} (A \cos 3t + B \sin 3t + \sin t)] \\ &= -2e^{-2t} (A \cos 3t + B \sin 3t + \sin t) \\ &\quad + e^{-2t} (-3A \sin 3t + 3B \cos 3t + \cos t) \\ &= e^{-2t} (-3A \sin 3t + 3B \cos 3t + \cos t - 2A \cos 3t - 2B \sin 3t - 2 \sin t) \end{aligned}$$

At  $t = 0$ ,

$$\frac{dy}{dt}(0) = 3B + 1 - 2A$$

Since  $\frac{dy}{dt}(0) = 2$  and  $A = 1$ ,

$$2 = 3B + 1 - 2$$

So

$$2 = 3B - 1$$

Hence

$$3B = 3$$

So

$$B = 1$$

Substituting  $A = 1$  and  $B = 1$  into the solution,

$$y = e^{-2t} (\cos 3t + \sin 3t + \sin t)$$

Hence the required particular solution is

$$y = e^{-2t} (\cos 3t + \sin 3t + \sin t)$$

3. A damped oscillator is governed by

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 13y = 75 \sin 2t$$

(a) Find the general solution of this differential equation. [6]

(b) Show that, whatever the initial conditions, the motion is approximately of the form

$$y \approx R \sin(2t - \phi)$$

for large positive values of  $t$ , where  $R > 0$  and  $0 < \phi < \frac{\pi}{2}$ . Determine  $R$  and  $\phi$ . [4]

### Solution

(a) First solve the associated homogeneous equation

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 13y = 0$$

Using the trial solution  $y = e^{\lambda t}$  gives the auxiliary equation

$$\lambda^2 + 6\lambda + 13 = 0$$

Solving,

$$\begin{aligned}\lambda &= \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2} \\ &= -3 \pm 2i\end{aligned}$$

Hence the complementary function is

$$y_c = e^{-3t}(A \cos 2t + B \sin 2t)$$

For a particular integral, try

$$y_p = a \sin 2t + b \cos 2t$$

Then

$$\begin{aligned}\frac{dy_p}{dt} &= 2a \cos 2t - 2b \sin 2t \\ \frac{d^2y_p}{dt^2} &= -4a \sin 2t - 4b \cos 2t\end{aligned}$$

Substituting into the differential equation:

$$\begin{aligned}(-4a \sin 2t - 4b \cos 2t) + 6(2a \cos 2t - 2b \sin 2t) + 13(a \sin 2t + b \cos 2t) \\ = 75 \sin 2t\end{aligned}$$

Collecting  $\sin 2t$  and  $\cos 2t$  terms,

$$\begin{aligned}(-4a - 12b + 13a) \sin 2t + (12a - 4b + 13b) \cos 2t \\ = 75 \sin 2t\end{aligned}$$

So

$$(9a - 12b) \sin 2t + (12a + 9b) \cos 2t = 75 \sin 2t$$

Equating coefficients gives

$$9a - 12b = 75$$

$$12a + 9b = 0$$

Solve these simultaneously. From the second equation,

$$4a + 3b = 0 \Rightarrow b = -\frac{4a}{3}$$

Substitute into the first:

$$9a - 12\left(-\frac{4a}{3}\right) = 75$$

$$9a + 16a = 75$$

$$25a = 75$$

$$a = 3$$

Then

$$b = -\frac{4(3)}{3} = -4$$

Thus

$$y_p = 3 \sin 2t - 4 \cos 2t$$

Therefore the general solution is

$$y = e^{-3t}(A \cos 2t + B \sin 2t) + 3 \sin 2t - 4 \cos 2t$$

(b) From part (a),

$$y = e^{-3t}(A \cos 2t + B \sin 2t) + 3 \sin 2t - 4 \cos 2t$$

As  $t \rightarrow +\infty$ , the exponential factor tends to 0. Also  $\sin 2t$  and  $\cos 2t$  remain bounded, so

$$e^{-3t}(A \cos 2t + B \sin 2t) \rightarrow 0$$

Therefore, whatever the initial conditions,

$$y \approx 3 \sin 2t - 4 \cos 2t$$

for large positive  $t$ .

Now write this in the form  $R \sin(2t - \phi)$ . Using the identity

$$\sin(2t - \phi) = \sin 2t \cos \phi - \cos 2t \sin \phi$$

we have

$$R \sin(2t - \phi) = R \cos \phi \sin 2t - R \sin \phi \cos 2t$$

Comparing with  $3 \sin 2t - 4 \cos 2t$  gives

$$R \cos \phi = 3$$

$$R \sin \phi = 4$$

Squaring and adding,

$$R^2 \cos^2 \phi + R^2 \sin^2 \phi = 3^2 + 4^2$$

$$R^2(\cos^2 \phi + \sin^2 \phi) = 25$$

$$R^2 = 25$$

Since  $R > 0$ ,

$$R = 5$$

Also,

$$\tan \phi = \frac{R \sin \phi}{R \cos \phi} = \frac{4}{3}$$

Since  $0 < \phi < \frac{\pi}{2}$ ,

$$\phi = \arctan \left( \frac{4}{3} \right) \approx 0.927$$

Hence for large positive  $t$ ,

$$y \approx 5 \sin \left( 2t - \arctan \frac{4}{3} \right)$$

So the required constants are  $R = 5$  and  $\phi = \arctan \left( \frac{4}{3} \right) \approx 0.927$

4. A function  $y(t)$  satisfies the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 10 \cos t$$

Given that, when  $t = 0$ ,  $y = 5$  and  $\frac{dy}{dt} = 2$ , find the particular solution.

[10]

### Solution

First solve the associated homogeneous equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 2\lambda + 5 = 0$$

Solving,

$$\begin{aligned}\lambda &= \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2} \\ &= -1 \pm 2i\end{aligned}$$

So the complementary function is

$$y_c = e^{-t}(C_1 \cos 2t + C_2 \sin 2t)$$

Now find a particular integral for

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 10 \cos t$$

Since the forcing term is  $10 \cos t$ , try

$$y_p = A \cos t + B \sin t$$

Then

$$\frac{dy_p}{dt} = -A \sin t + B \cos t$$

and

$$\frac{d^2y_p}{dt^2} = -A \cos t - B \sin t$$

Substituting into the differential equation,

$$(-A \cos t - B \sin t) + 2(-A \sin t + B \cos t) + 5(A \cos t + B \sin t) = 10 \cos t$$

Collecting  $\cos t$  and  $\sin t$  terms,

$$\begin{aligned}(-A + 2B + 5A) \cos t + (-B - 2A + 5B) \sin t &= 10 \cos t \\ (4A + 2B) \cos t + (4B - 2A) \sin t &= 10 \cos t\end{aligned}$$

Equating coefficients gives

$$4A + 2B = 10, \quad 4B - 2A = 0$$

From  $4B - 2A = 0$ ,

$$A = 2B$$

Substitute into  $4A + 2B = 10$ :

$$\begin{aligned}4(2B) + 2B &= 10 \\ 8B + 2B &= 10 \\ 10B &= 10 \\ B &= 1\end{aligned}$$

Hence

$$A = 2$$

So the particular integral is

$$y_p = 2 \cos t + \sin t$$

Therefore the general solution is

$$y = e^{-t}(C_1 \cos 2t + C_2 \sin 2t) + 2 \cos t + \sin t$$

Now use the initial conditions.

When  $t = 0$ ,  $y = 5$ :

$$\begin{aligned} 5 &= e^0(C_1 \cos 0 + C_2 \sin 0) + 2 \cos 0 + \sin 0 \\ &= C_1 + 2 \end{aligned}$$

So

$$C_1 = 3$$

Next differentiate the general solution:

$$\frac{dy}{dt} = \frac{d}{dt} [e^{-t}(C_1 \cos 2t + C_2 \sin 2t)] + \frac{d}{dt}(2 \cos t + \sin t)$$

Using the product rule on the first term,

$$\begin{aligned} \frac{dy}{dt} &= -e^{-t}(C_1 \cos 2t + C_2 \sin 2t) + e^{-t}(-2C_1 \sin 2t + 2C_2 \cos 2t) - 2 \sin t + \cos t \\ &= e^{-t} [(-C_1 + 2C_2) \cos 2t + (-2C_1 - C_2) \sin 2t] - 2 \sin t + \cos t \end{aligned}$$

Now use  $\frac{dy}{dt} = 2$  when  $t = 0$ :

$$\begin{aligned} 2 &= e^0 [(-C_1 + 2C_2) \cos 0 + (-2C_1 - C_2) \sin 0] - 2 \sin 0 + \cos 0 \\ &= (-C_1 + 2C_2) + 1 \end{aligned}$$

Substitute  $C_1 = 3$ :

$$2 = -3 + 2C_2 + 1$$

$$2 = 2C_2 - 2$$

$$4 = 2C_2$$

$$C_2 = 2$$

Hence the required particular solution satisfying the given conditions is

$$y(t) = e^{-t}(3 \cos 2t + 2 \sin 2t) + 2 \cos t + \sin t$$

5. (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0 \quad [2]$$

(b) Hence find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = e^{-2x}(6 \cos x - 8 \sin x) \quad [4]$$

### Solution

(a) For the homogeneous equation, use the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

Using the quadratic formula,

$$\begin{aligned} \lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} \\ &= -2 \pm 3i \end{aligned}$$

Hence the complementary solution is

$$y = e^{-2x} (A \cos 3x + B \sin 3x)$$

So the general solution is  $y = e^{-2x} (A \cos 3x + B \sin 3x)$

(b) From part (a), the complementary function is

$$y_c = e^{-2x} (C \cos 3x + D \sin 3x)$$

For a particular integral, try

$$y_p = e^{-2x} (a \cos x + b \sin x)$$

Then

$$\begin{aligned} \frac{dy_p}{dx} &= \frac{d}{dx} [e^{-2x} (a \cos x + b \sin x)] \\ &= -2e^{-2x} (a \cos x + b \sin x) + e^{-2x} (-a \sin x + b \cos x) \\ &= e^{-2x} [(b - 2a) \cos x + (-a - 2b) \sin x] \end{aligned}$$

Also,

$$\begin{aligned} \frac{d^2y_p}{dx^2} &= \frac{d}{dx} \{e^{-2x} [(b - 2a) \cos x + (-a - 2b) \sin x]\} \\ &= -2e^{-2x} [(b - 2a) \cos x + (-a - 2b) \sin x] \\ &\quad + e^{-2x} [(-b + 2a) \sin x + (-a - 2b) \cos x] \\ &= e^{-2x} [(3a - 4b) \cos x + (4a + 3b) \sin x] \end{aligned}$$

Substituting into

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = e^{-2x}(6 \cos x - 8 \sin x)$$

gives

$$\begin{aligned} & e^{-2x} [(3a - 4b) \cos x + (4a + 3b) \sin x] \\ & + 4e^{-2x} [(b - 2a) \cos x + (-a - 2b) \sin x] \\ & + 13e^{-2x} (a \cos x + b \sin x) \\ & = e^{-2x} (6 \cos x - 8 \sin x) \end{aligned}$$

So

$$e^{-2x} (8a \cos x + 8b \sin x) = e^{-2x} (6 \cos x - 8 \sin x)$$

Hence

$$8a \cos x + 8b \sin x = 6 \cos x - 8 \sin x$$

Comparing coefficients,

$$8a = 6, \quad 8b = -8$$

So

$$a = \frac{3}{4}, \quad b = -1$$

Therefore

$$y_p = e^{-2x} \left( \frac{3}{4} \cos x - \sin x \right)$$

Thus the general solution is

$$y = e^{-2x} (C \cos 3x + D \sin 3x) + e^{-2x} \left( \frac{3}{4} \cos x - \sin x \right)$$

So

$$y = e^{-2x} \left( C \cos 3x + D \sin 3x + \frac{3}{4} \cos x - \sin x \right)$$

6. (a) Show that

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

[2]

(b) Hence find the particular solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4\left(\frac{\cosh x - \sinh x}{\cosh x + \sinh x}\right)$$

given that, when  $x = 0$ ,  $y = 1$  and  $\frac{dy}{dx} = 1$ .

[10]

### Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

we get

$$\begin{aligned} \cosh x - \sinh x &= \frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^{-x}}{2} \\ &= e^{-x} \end{aligned}$$

Also,

$$\begin{aligned} \cosh x + \sinh x &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} + e^x - e^{-x}}{2} \\ &= \frac{2e^x}{2} \\ &= e^x \end{aligned}$$

Therefore,

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = \frac{e^{-x}}{e^x} = e^{-2x}$$

Hence,

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

(b) From part (a), the differential equation becomes

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4e^{-2x}$$

For the homogeneous equation,

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 4 = 0$$

So

$$(\lambda + 2)^2 = 0$$

Hence the complementary function is

$$y_c = (Cx + D)e^{-2x}$$

Since the right-hand side is  $4e^{-2x}$  and  $-2$  is a repeated root of the auxiliary equation, try

$$y_p = Ax^2e^{-2x}$$

Differentiate:

$$\begin{aligned}\frac{dy_p}{dx} &= \frac{d}{dx} (Ax^2e^{-2x}) \\ &= 2Axe^{-2x} - 2Ax^2e^{-2x} \\ &= e^{-2x}(2Ax - 2Ax^2)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{d^2y_p}{dx^2} &= \frac{d}{dx} [e^{-2x}(2Ax - 2Ax^2)] \\ &= -2e^{-2x}(2Ax - 2Ax^2) + e^{-2x}(2A - 4Ax) \\ &= e^{-2x}(2A - 8Ax + 4Ax^2)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{d^2y_p}{dx^2} + 4\frac{dy_p}{dx} + 4y_p &= e^{-2x}(2A - 8Ax + 4Ax^2) + 4e^{-2x}(2Ax - 2Ax^2) + 4Ax^2e^{-2x} \\ &= e^{-2x}(2A)\end{aligned}$$

So

$$2Ae^{-2x} = 4e^{-2x}$$

Hence

$$2A = 4$$

So

$$A = 2$$

Therefore

$$y_p = 2x^2e^{-2x}$$

Thus the general solution is

$$y = (Cx + D)e^{-2x} + 2x^2e^{-2x}$$

So

$$y = e^{-2x}(2x^2 + Cx + D)$$

Now use the initial conditions

Since  $y(0) = 1$ ,

$$\begin{aligned}1 &= e^0(2 \cdot 0^2 + C \cdot 0 + D) \\ &= D\end{aligned}$$

So

$$D = 1$$

Now differentiate

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [e^{-2x}(2x^2 + Cx + D)] \\ &= -2e^{-2x}(2x^2 + Cx + D) + e^{-2x}(4x + C) \\ &= e^{-2x}(4x + C - 4x^2 - 2Cx - 2D)\end{aligned}$$

At  $x = 0$ ,

$$\frac{dy}{dx}(0) = C - 2D$$

Since  $\frac{dy}{dx}(0) = 1$  and  $D = 1$ ,

$$1 = C - 2$$

Hence

$$C = 3$$

Substituting  $C = 3$  and  $D = 1$  into the solution gives

$$y = (2x^2 + 3x + 1)e^{-2x}$$

Therefore the required particular solution is

$$y = (2x^2 + 3x + 1)e^{-2x}$$

7. Solve the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 50 \sin x$$

subject to the conditions  $y(0) = 3$  and  $y'(0) = -6$ .

[10]

### Solution

We solve

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 50 \sin x$$

subject to

$$y(0) = 3, \quad y'(0) = -6$$

First, solve the homogeneous equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$$

Its auxiliary equation is

$$\lambda^2 - \lambda - 6 = 0$$

Factorising,

$$(\lambda - 3)(\lambda + 2) = 0$$

So the roots are  $\lambda = 3$  and  $\lambda = -2$ . Hence the complementary function is

$$y_c = Ae^{3x} + Be^{-2x}$$

Now find a particular integral. Since the right-hand side is  $50 \sin x$ , try

$$y_p = a \sin x + b \cos x$$

Then

$$\begin{aligned} \frac{dy_p}{dx} &= a \cos x - b \sin x \\ \frac{d^2y_p}{dx^2} &= -a \sin x - b \cos x \end{aligned}$$

Substituting into the differential equation,

$$\begin{aligned} \frac{d^2y_p}{dx^2} - \frac{dy_p}{dx} - 6y_p &= (-a \sin x - b \cos x) - (a \cos x - b \sin x) - 6(a \sin x + b \cos x) \\ &= -a \sin x - b \cos x - a \cos x + b \sin x - 6a \sin x - 6b \cos x \\ &= (b - 7a) \sin x + (-a - 7b) \cos x \end{aligned}$$

We want this to equal  $50 \sin x$ , so equating coefficients gives

$$\begin{aligned} b - 7a &= 50 \\ -a - 7b &= 0 \end{aligned}$$

From the second equation,

$$a = -7b$$

Substitute into the first:

$$\begin{aligned} b - 7(-7b) &= 50 \\ b + 49b &= 50 \\ 50b &= 50 \\ b &= 1 \end{aligned}$$

So

$$a = -7$$

Therefore the particular integral is

$$y_p = -7 \sin x + \cos x$$

Hence the general solution is

$$y = Ae^{3x} + Be^{-2x} - 7 \sin x + \cos x$$

Now use the initial conditions. First,

$$\begin{aligned} y(0) &= Ae^0 + Be^0 - 7 \sin 0 + \cos 0 \\ &= A + B + 1 \end{aligned}$$

Since  $y(0) = 3$ ,

$$A + B + 1 = 3$$

so

$$A + B = 2$$

Next differentiate the general solution:

$$y'(x) = 3Ae^{3x} - 2Be^{-2x} - 7 \cos x - \sin x$$

Now apply  $y'(0) = -6$ :

$$\begin{aligned} y'(0) &= 3Ae^0 - 2Be^0 - 7 \cos 0 - \sin 0 \\ &= 3A - 2B - 7 \end{aligned}$$

So

$$3A - 2B - 7 = -6$$

hence

$$3A - 2B = 1$$

We now solve

$$\begin{aligned} A + B &= 2 \\ 3A - 2B &= 1 \end{aligned}$$

Multiply the first equation by 2:

$$2A + 2B = 4$$

Add this to  $3A - 2B = 1$ :

$$\begin{aligned} (2A + 2B) + (3A - 2B) &= 4 + 1 \\ 5A &= 5 \\ A &= 1 \end{aligned}$$

Then from  $A + B = 2$ ,

$$\begin{aligned} 1 + B &= 2 \\ B &= 1 \end{aligned}$$

Therefore the required solution is

$$y = e^{3x} + e^{-2x} - 7 \sin x + \cos x$$

8. Determine the solution  $y(x)$  of the initial value problem

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = xe^{2x}$$

subject to

$$y(0) = 1, \quad \left. \frac{dy}{dx} \right|_{x=0} = 0 \quad [10]$$

### Solution

For the homogeneous equation,

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 - 4\lambda + 4 = 0$$

So

$$(\lambda - 2)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^{2x}$$

For a particular integral, since the right-hand side is  $xe^{2x}$  and  $\lambda = 2$  is a repeated root, try

$$y_p = e^{2x}(ax^3 + bx^2)$$

Then

$$\begin{aligned} \frac{dy_p}{dx} &= \frac{d}{dx} [e^{2x}(ax^3 + bx^2)] \\ &= 2e^{2x}(ax^3 + bx^2) + e^{2x}(3ax^2 + 2bx) \\ &= e^{2x}(2ax^3 + (2b + 3a)x^2 + 2bx) \end{aligned}$$

Differentiate again:

$$\begin{aligned} \frac{d^2y_p}{dx^2} &= \frac{d}{dx} [e^{2x}(2ax^3 + (2b + 3a)x^2 + 2bx)] \\ &= 2e^{2x}(2ax^3 + (2b + 3a)x^2 + 2bx) \\ &\quad + e^{2x}(6ax^2 + 2(2b + 3a)x + 2b) \\ &= e^{2x}(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b) \end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned} \frac{d^2y_p}{dx^2} - 4\frac{dy_p}{dx} + 4y_p &= e^{2x}(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b) \\ &\quad - 4e^{2x}(2ax^3 + (2b + 3a)x^2 + 2bx) \\ &\quad + 4e^{2x}(ax^3 + bx^2) \end{aligned}$$

So

$$\frac{d^2y_p}{dx^2} - 4\frac{dy_p}{dx} + 4y_p = e^{2x}(6ax + 2b)$$

Comparing with  $xe^{2x}$ , we get

$$6a = 1, \quad 2b = 0$$

So

$$a = \frac{1}{6}, \quad b = 0$$

Therefore

$$y_p = \frac{x^3}{6}e^{2x}$$

Thus the general solution is

$$y = (A + Bx)e^{2x} + \frac{x^3}{6}e^{2x}$$

So

$$y = e^{2x} \left( A + Bx + \frac{x^3}{6} \right)$$

Now use the initial conditions

From  $y(0) = 1$ ,

$$\begin{aligned} 1 &= e^0 \left( A + B \cdot 0 + \frac{0^3}{6} \right) \\ &= A \end{aligned}$$

So

$$A = 1$$

Now differentiate

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \left[ e^{2x} \left( A + Bx + \frac{x^3}{6} \right) \right] \\ &= 2e^{2x} \left( A + Bx + \frac{x^3}{6} \right) + e^{2x} \left( B + \frac{x^2}{2} \right) \\ &= e^{2x} \left( 2A + 2Bx + \frac{x^3}{3} + B + \frac{x^2}{2} \right) \end{aligned}$$

At  $x = 0$ ,

$$\frac{dy}{dx}(0) = 2A + B$$

Since  $\frac{dy}{dx}(0) = 0$  and  $A = 1$ ,

$$0 = 2 + B$$

Hence

$$B = -2$$

Substituting  $A = 1$  and  $B = -2$  gives

$$y(x) = e^{2x} \left( \frac{x^3}{6} - 2x + 1 \right)$$

Therefore, the solution of the initial value problem is

$$y(x) = e^{2x} \left( \frac{x^3}{6} - 2x + 1 \right)$$

9. (a) Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 0 \quad [2]$$

- (b) Hence, by considering a particular integral of the form  $x(A \cos 2x + B \sin 2x)$ , find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 8 \cos 2x - 12 \sin 2x \quad [4]$$

### Solution

- (a) For

$$\frac{d^2y}{dx^2} + 4y = 0$$

assume a solution of the form  $y = e^{\lambda x}$ . Then the auxiliary equation is

$$\lambda^2 + 4 = 0$$

So

$$\lambda = \pm 2i$$

Hence the general solution is

$$y = C_1 \cos 2x + C_2 \sin 2x$$

- (b) From part (a), the complementary solution is

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

Since the right-hand side involves  $\cos 2x$  and  $\sin 2x$ , take a particular solution in the given form

$$y_p = x(A \cos 2x + B \sin 2x)$$

Differentiate:

$$\frac{dy_p}{dx} = (A \cos 2x + B \sin 2x) + x(-2A \sin 2x + 2B \cos 2x)$$

Differentiate again:

$$\begin{aligned} \frac{d^2y_p}{dx^2} &= (-2A \sin 2x + 2B \cos 2x) + (-2A \sin 2x + 2B \cos 2x) \\ &\quad + x(-4A \cos 2x - 4B \sin 2x) \\ &= -4A \sin 2x + 4B \cos 2x - 4x(A \cos 2x + B \sin 2x) \end{aligned}$$

Now substitute into  $\frac{d^2y}{dx^2} + 4y$ :

$$\begin{aligned} \frac{d^2y_p}{dx^2} + 4y_p &= [-4A \sin 2x + 4B \cos 2x - 4x(A \cos 2x + B \sin 2x)] \\ &\quad + 4x(A \cos 2x + B \sin 2x) \\ &= 4B \cos 2x - 4A \sin 2x \end{aligned}$$

This must equal the right-hand side:

$$4B \cos 2x - 4A \sin 2x = 8 \cos 2x - 12 \sin 2x$$

Equating coefficients gives

$$\begin{aligned} 4B &= 8 \\ -4A &= -12 \end{aligned}$$

so

$$B = 2, \quad A = 3$$

Therefore

$$y_p = x(3 \cos 2x + 2 \sin 2x)$$

Hence the general solution is

$$y = y_c + y_p = C_1 \cos 2x + C_2 \sin 2x + x(3 \cos 2x + 2 \sin 2x)$$

10. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 28 \cos 3t - 4 \sin 3t$$

given that, when  $t = 0$ ,  $y = 5$  and  $\frac{dy}{dt} = 7$

[10]

### Solution

We solve

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 28 \cos 3t - 4 \sin 3t$$

by finding the complementary function and a particular integral, then using the initial conditions. For the complementary function, solve the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

Using the quadratic formula,

$$\begin{aligned}\lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} \\ &= -2 \pm 3i\end{aligned}$$

Hence the complementary function is

$$y_c = e^{-2t}(A \cos 3t + B \sin 3t)$$

For the particular integral, since the right-hand side is of the form  $28 \cos 3t - 4 \sin 3t$ , try

$$y_p = a \cos 3t + b \sin 3t$$

Then

$$\begin{aligned}\frac{dy_p}{dt} &= -3a \sin 3t + 3b \cos 3t \\ \frac{d^2y_p}{dt^2} &= -9a \cos 3t - 9b \sin 3t\end{aligned}$$

Substituting into the differential equation,

$$\begin{aligned}\frac{d^2y_p}{dt^2} + 4\frac{dy_p}{dt} + 13y_p &= (-9a \cos 3t - 9b \sin 3t) + 4(-3a \sin 3t + 3b \cos 3t) + 13(a \cos 3t + b \sin 3t) \\ &= (-9a + 12b + 13a) \cos 3t + (-9b - 12a + 13b) \sin 3t \\ &= (4a + 12b) \cos 3t + (4b - 12a) \sin 3t\end{aligned}$$

Equating coefficients with  $28 \cos 3t - 4 \sin 3t$  gives

$$\begin{aligned}4a + 12b &= 28 \\ 4b - 12a &= -4\end{aligned}$$

Dividing both equations by 4,

$$\begin{aligned}a + 3b &= 7 \\ b - 3a &= -1\end{aligned}$$

From the second equation,

$$b = 3a - 1$$

Substitute into the first:

$$\begin{aligned}a + 3(3a - 1) &= 7 \\a + 9a - 3 &= 7 \\10a &= 10 \\a &= 1\end{aligned}$$

Then

$$b = 3(1) - 1 = 2$$

So

$$y_p = \cos 3t + 2 \sin 3t$$

Hence the general solution is

$$y = e^{-2t}(A \cos 3t + B \sin 3t) + \cos 3t + 2 \sin 3t$$

Now use the initial conditions.

First, when  $t = 0$ ,  $y = 5$ :

$$\begin{aligned}5 &= e^0(A \cos 0 + B \sin 0) + \cos 0 + 2 \sin 0 \\&= A(1) + B(0) + 1 + 0 \\&= A + 1\end{aligned}$$

So

$$A = 4$$

Next differentiate  $y$ .

For the homogeneous part,

$$\begin{aligned}\frac{d}{dt} [e^{-2t}(A \cos 3t + B \sin 3t)] &= e^{-2t}(-2)(A \cos 3t + B \sin 3t) + e^{-2t}(-3A \sin 3t + 3B \cos 3t) \\&= e^{-2t}[(3B - 2A) \cos 3t + (-3A - 2B) \sin 3t]\end{aligned}$$

Also,

$$\frac{d}{dt} (\cos 3t + 2 \sin 3t) = -3 \sin 3t + 6 \cos 3t$$

Therefore

$$\frac{dy}{dt} = e^{-2t}[(3B - 2A) \cos 3t + (-3A - 2B) \sin 3t] - 3 \sin 3t + 6 \cos 3t$$

Using  $\frac{dy}{dt} = 7$  when  $t = 0$ ,

$$\begin{aligned}7 &= e^0[(3B - 2A) \cos 0 + (-3A - 2B) \sin 0] - 3 \sin 0 + 6 \cos 0 \\&= (3B - 2A)(1) + 0 + 0 + 6 \\&= 3B - 2A + 6\end{aligned}$$

Substitute  $A = 4$ :

$$\begin{aligned}7 &= 3B - 8 + 6 \\7 &= 3B - 2 \\3B &= 9 \\B &= 3\end{aligned}$$

Hence the required particular solution is

$$y = (4 \cos 3t + 3 \sin 3t)e^{-2t} + \cos 3t + 2 \sin 3t$$

11. Find the particular solution of the differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = 8\cos t$$

given that, when  $t = 0$ ,  $y = 3$  and  $\frac{dy}{dt} = 1$

[10]

### Solution

First solve the associated homogeneous equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 4\lambda + 5 = 0$$

Using the quadratic formula:

$$\begin{aligned}\lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)} \\ &= -2 \pm i\end{aligned}$$

Hence the complementary function is

$$y_c = e^{-2t}(C_1 \cos t + C_2 \sin t)$$

Now find a particular integral for

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = 8\cos t$$

Since the forcing term is  $8\cos t$ , try

$$y_p = A \cos t + B \sin t$$

Then

$$\frac{dy_p}{dt} = -A \sin t + B \cos t$$

and

$$\frac{d^2y_p}{dt^2} = -A \cos t - B \sin t$$

Substituting into the differential equation:

$$\begin{aligned}\frac{d^2y_p}{dt^2} + 4\frac{dy_p}{dt} + 5y_p &= (-A \cos t - B \sin t) + 4(-A \sin t + B \cos t) + 5(A \cos t + B \sin t) \\ &= (-A + 4B + 5A) \cos t + (-B - 4A + 5B) \sin t \\ &= (4A + 4B) \cos t + (-4A + 4B) \sin t\end{aligned}$$

This must equal  $8\cos t$ , so equating coefficients gives

$$4A + 4B = 8, \quad -4A + 4B = 0$$

Divide both equations by 4:

$$A + B = 2, \quad -A + B = 0$$

From  $-A + B = 0$ , we get

$$B = A$$

Substitute into  $A + B = 2$ :

$$A + A = 2$$

$$2A = 2$$

$$A = 1$$

Therefore

$$B = 1$$

So the particular integral is

$$y_p = \cos t + \sin t$$

Hence the general solution is

$$y = e^{-2t}(C_1 \cos t + C_2 \sin t) + \cos t + \sin t$$

Now use the initial conditions.

Given  $y(0) = 3$ , substitute  $t = 0$ :

$$\begin{aligned} 3 &= e^0(C_1 \cos 0 + C_2 \sin 0) + \cos 0 + \sin 0 \\ &= 1(C_1 \cdot 1 + C_2 \cdot 0) + 1 + 0 \\ &= C_1 + 1 \end{aligned}$$

So

$$C_1 + 1 = 3$$

$$C_1 = 2$$

Next differentiate the general solution:

$$\begin{aligned} \frac{dy}{dt} &= \frac{d}{dt} [e^{-2t}(C_1 \cos t + C_2 \sin t)] + \frac{d}{dt} (\cos t + \sin t) \\ &= -2e^{-2t}(C_1 \cos t + C_2 \sin t) + e^{-2t}(-C_1 \sin t + C_2 \cos t) - \sin t + \cos t \\ &= e^{-2t} [(-2C_1 + C_2) \cos t + (-C_1 - 2C_2) \sin t] - \sin t + \cos t \end{aligned}$$

Given  $\frac{dy}{dt}(0) = 1$ , substitute  $t = 0$ :

$$\begin{aligned} 1 &= e^0 [(-2C_1 + C_2) \cos 0 + (-C_1 - 2C_2) \sin 0] - \sin 0 + \cos 0 \\ &= 1 [(-2C_1 + C_2) \cdot 1 + (-C_1 - 2C_2) \cdot 0] - 0 + 1 \\ &= (-2C_1 + C_2) + 1 \end{aligned}$$

So

$$(-2C_1 + C_2) + 1 = 1$$

$$-2C_1 + C_2 = 0$$

Using  $C_1 = 2$ :

$$-2(2) + C_2 = 0$$

$$-4 + C_2 = 0$$

$$C_2 = 4$$

Therefore the particular solution satisfying the given conditions is

$$y = e^{-2t}(2 \cos t + 4 \sin t) + \cos t + \sin t$$

12. (a) Show that, for all real  $x$ ,

$$(1 + \tanh x) \cosh x = e^x \quad [2]$$

(b) Hence solve the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 3(1 + \tanh x) \cosh x$$

given that, when  $x = 0$ ,  $y = 1$  and  $\frac{dy}{dx} = 0$

[10]

### Solution

(a) Using  $\tanh x = \frac{\sinh x}{\cosh x}$ ,

$$\begin{aligned} (1 + \tanh x) \cosh x &= \left(1 + \frac{\sinh x}{\cosh x}\right) \cosh x \\ &= \cosh x + \sinh x \\ &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= e^x \end{aligned}$$

Hence, for all real  $x$ ,

$$(1 + \tanh x) \cosh x = e^x$$

(b) From part (a), the differential equation becomes

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 3e^x$$

For the homogeneous equation,

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$$

The auxiliary equation is

$$\lambda^2 - 2\lambda + 1 = 0$$

So

$$(\lambda - 1)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^x$$

For a particular integral, since the right-hand side is  $3e^x$  and  $\lambda = 1$  is a repeated root, try

$$y_p = Cx^2e^x$$

Then

$$\begin{aligned} \frac{dy_p}{dx} &= \frac{d}{dx} (Cx^2e^x) \\ &= 2Cxe^x + Cx^2e^x \\ &= e^x (Cx^2 + 2Cx) \end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{d^2 y_p}{dx^2} &= \frac{d}{dx} [e^x (Cx^2 + 2Cx)] \\ &= e^x (Cx^2 + 2Cx) + e^x (2Cx + 2C) \\ &= e^x (Cx^2 + 4Cx + 2C)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{d^2 y_p}{dx^2} - 2 \frac{dy_p}{dx} + y_p &= e^x (Cx^2 + 4Cx + 2C) - 2e^x (Cx^2 + 2Cx) + Cx^2 e^x \\ &= 2Ce^x\end{aligned}$$

So

$$2Ce^x = 3e^x$$

Hence

$$2C = 3$$

So

$$C = \frac{3}{2}$$

Therefore

$$y_p = \frac{3x^2}{2} e^x$$

Thus the general solution is

$$y = (A + Bx)e^x + \frac{3x^2}{2} e^x$$

So

$$y = e^x \left( A + Bx + \frac{3x^2}{2} \right)$$

Now use the initial conditions

From  $y(0) = 1$ ,

$$\begin{aligned}1 &= e^0 \left( A + B \cdot 0 + \frac{3(0)^2}{2} \right) \\ &= A\end{aligned}$$

So

$$A = 1$$

Now differentiate

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left[ e^x \left( A + Bx + \frac{3x^2}{2} \right) \right] \\ &= e^x \left( A + Bx + \frac{3x^2}{2} \right) + e^x (B + 3x) \\ &= e^x \left( A + Bx + \frac{3x^2}{2} + B + 3x \right)\end{aligned}$$

At  $x = 0$ ,

$$\frac{dy}{dx}(0) = A + B$$

Since  $\frac{dy}{dx}(0) = 0$  and  $A = 1$ ,

$$0 = 1 + B$$

Hence

$$B = -1$$

Substituting  $A = 1$  and  $B = -1$  into the expression for  $y$ ,

$$y = e^x \left( \frac{3x^2}{2} - x + 1 \right)$$

Therefore the required solution is

$$y = e^x \left( \frac{3x^2}{2} - x + 1 \right)$$